

MOHAMED AMINE ELKADHI

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PROFESSIONAL SUMMARY

Industrial Computer Science and Automation engineering student (INSAT), also pursuing a Research Master’s in IoT and Data Analytics (École Polytechnique de Tunisie), with strong interests in manufacturing systems optimization, OEE/TRS analysis, and industrial automation. Led IEEE RAS INSAT to Best Student Branch Chapter Worldwide. Seeking an end-of-study internship (PFE) for my Engineering Degree in Industrial Computer Science and Automation.

EDUCATION

National Institute of Applied Science and Technology (INSAT), Tunis, Tunisia — Engineering Degree, Industrial Computer Science and Automation | Sep 2022 – Sep 2027

École Polytechnique de Tunisie — Research Master’s Degree, IoT and Data Analytics | Sep 2026 – Sep 2027

PROFESSIONAL EXPERIENCE

Opella (Sanofi Consumer Healthcare) — Industrial Engineering Intern Jul – Sep 2026

- Decision-support dashboard (PDSS-CTA): designed and built, in Python/Streamlit, a modular prescriptive maintenance diagnostic tool for Air Handling Units, structured as a 3-stage cascading elimination process (subsystem → technical fault → root cause) driven by a single reusable questionnaire engine shared across all three diagnostic levels.
- Implemented a weighted, multi-field fuzzy-matching engine (rapidfuzz) that scores free-text symptom queries against a 45-rule maintenance knowledge base spanning 10 subsystems, with an analogical-inference fallback for unrecognized symptoms and an auto-generated downloadable diagnostic report.
- OEE/TRS production dashboard: analyzing Opella’s 5-year (2025–2030) production plan across 4 production lines to build a Power BI dashboard computing global and per-machine TRS/OEE, including availability-rate (TO) calculations.
- Modeling available monthly and yearly production capacity per product and calculating the time required to reach production targets under 3-, 4-, and 5-shift scenarios, to support capacity-planning decisions.

Ulysse Logistics Company (UTIC Group) — Logistics Systems Optimization Intern Jul – Aug 2025

- Conducted on-site Gemba walk analysis of logistics flows and order-preparation processes; analyzed historical pallet-volume and store-planning data to design a dynamic loading-dock allocation model.
- Reorganized preparation areas and integrated the allocation model into the operational system (Radio Pad), cutting dock occupancy imbalance by 40% and improving flow efficiency and safety.

TECHNICAL PROJECTS

- EUROBOT Senior 2026 – Navigation Engineer, IEEE RAS INSAT SBC:** developed odometry, PID control, and interrupt-based motion handling (C++, Arduino IDE, ESP32) for an autonomous robot completing tasks within 100 seconds; qualified for the EUROBOT 2026 International Finals (France).
- AnyLogic Manufacturing Supply Chain Simulation:** modeled and evaluated a manufacturing supply chain with custom Java simulation logic; used Excel for KPI analysis, scenario testing, and bottleneck identification.
- Virtual Industry and Automation Simulation:** end-to-end production automation simulation (TIA Portal, Factory I/O, GRAFCET, GEMMA) covering automation, inventory management, and quality control.
- Supply Chain Optimization & Operations Research:** modeled production planning, transportation, and product-mix problems using LP/MILP with sensitivity analysis for operational decision support.

TECHNICAL SKILLS

Manufacturing Engineering & Supply Chain: OEE/TRS Analysis, Lean Manufacturing, Six Sigma, Process Mapping, ERP/MRP (SAP), Supply Chain Management, LP/MILP Modeling

Data & BI: Power BI, Advanced Excel, SQL, Python (Pandas, Streamlit), Java

Automation & Embedded Systems: TIA Portal, GRAFCET, GEMMA, PLCs, C/C++, Arduino, STM32, ESP32, PID Control

Simulation & Modeling: AnyLogic, SolidWorks (3D)

Certifications: SAP & ERP/MRP Systems (WYNSYS); Embedded C, STM32 & Embedded Linux (CSF); Lean Manufacturing Green Belt – TU Munich (in progress)

Languages: Arabic (Native), French (Fluent), English (Fluent)

LEADERSHIP & AWARDS

Soft Skills: Leadership, Cross-Functional Communication, Problem-Solving, Team Management

IEEE RAS INSAT Student Branch Chapter – Chair (Aug 2024–Sep 2025): led 30+ members; grew the chapter to Best IEEE RAS Student Branch Chapter Worldwide (2024–2025) and Best RAS SBC in Tunisia (x2); introduced the NXP CUP to Tunisia.

IEEE INSAT Student Branch – Vice Chair, Technical Activities (2025–Present): directed all technical activities and mentored 9 teams at TSYP 13.0, leading the branch to overall victory and the Outstanding Student Branch Award (Large).

IEEE Region 8 MENA SYP Congress – Technical Manager; TSYP 14.0 – Grand Challenger Team Leader (2025)

Selected Awards: EUROBOT International Finals Qualifier (2025); 2nd Prize, IEEE IAS IPCSD Africa; 1st Place, AquaQuest Hackathon; Darrel Chong Silver Award (IEEE Foundation).